(*) All the rings should be regarded as commutative rings with identity if there is no special statement.

Algebra

Definition 0.1. :Let A, B be two rings and $f : A \to B$ is a ring homomorphism, define an A-module structure on B by the scalar multiplication

$$A \times B \to B$$

$$(a, b) \mapsto a \cdot b := f(a)b$$

then the A-module B is called an **A-algebra**.

Example 0.1. : Any ring A is a $\mathbb{Z}$-module via

$$\mathbb{Z} \to A$$

$$n \mapsto n \cdot 1$$

Example 0.2. : If A is a filed, then the ring homomorphism $f : A \to B$ is injective as long as f is not zero. So $A \cong f(A)$ can be viewed as a subring of B.

Example 0.3. : Let $B = A[x_1, x_2, \ldots, x_n]$, then B is an A-algebra via the inclusion homomorphism

$$A \hookrightarrow B = A[x_1, x_2, \ldots, x_n]$$

Definition 0.2. :(1) If B is an A-algebra, which is a finite A-module, then B is said to be a **finite A-algebra**.
(2) Let B be an A-algebra with the structure map $f : A \to B$, then B is said to be a **finitely generated A-algebra** if there exists an epimorphism $h : A[t_1, t_2, \ldots, t_n] \to B$. i.e., $\exists x_1, x_2, \ldots, x_n \in B$ such that for $\forall b \in B$, b is of the form

$$b = \sum_{i_1, i_2, \ldots, i_n} b_{i_1 i_2 \ldots i_n} x_1^{i_1} x_2^{i_2} \ldots x_n^{i_n},$$

where $b_{i_1 i_2 \ldots i_n} \in f(A)$.

Example 0.4. :(1) $\mathbb{Z}[x_1, x_2]/(x_1^2)$ is a finitely generated $\mathbb{Z}$-algebra but not a finite $\mathbb{Z}$-algebra since the $\mathbb{Z}$-basis is $\{\overline{1}, \overline{x}_1^i \overline{x}_2^j | \ 0 \leq i \leq 1, 0 \leq j < \infty\}$.

(2) $\mathbb{Z}[x]/(x^3)$ is a finite $\mathbb{Z}$-algebra but not a finitely generated $\mathbb{Z}$-algebra since we can know that $\{\overline{1}, \overline{x}, \overline{x}^2\}$ is the $\mathbb{Z}$-basis.

Proposition 0.1. : *If B, C are two A-algebras, then $B \otimes_A C$ is an A-module, define the multiplication on $B \otimes_A C$ as*

$$(b \otimes c) \cdot (b' \otimes c') := bb' \otimes cc'$$

then $B \otimes_A C$ becomes an A-algebra.

Proof: Straightforward vertification. $\square$

Multiplicative Set

Definition 0.3. : Let A be a ring, a subset $S \subseteq A$ is sai to be a **multiplicative set** if $1_A \in S$ and $a, b \in S \Rightarrow ab \in S$.

Now define a relation $\sim$ on $A \times S$ as:

$$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S \text{ such that } u(at - bs) = 0$$

then it can be checked that $\sim$ is actually an equivlence relation on $A \times S$. The set of all equivalence classes on $A \times S$ is denoted by $S^{-1}A$ and the equivalence class of (a, s) is denoted by $\frac{a}{s}$, i.e., $\frac{a}{s} = [(a, s)]$

$$S^{-1}A = \{ \frac{a}{s} \mid a \in A, s \in S\}.$$

Now define the addition and multiplcation on $S^{-1}A$ as

$$\frac{a}{s} + \frac{b}{t} := \frac{at + bs}{st}$$

$$\frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st}$$

it can be checked that such defined addition and multiplication are both well-defined, i.e., independent of the choice of the representative element.

It's easy to show that $S^{-1}A$ is a commutative ring with respect to the addition and multiplication above. The addition identity (i.e., the zero element) is $\frac{0}{1}$ and the multiplication identity is $\frac{1}{1}$.

Example 0.5. : $\mathbb{Q} = S^{-1}A$, where $S = \mathbb{Z} - \{0\}$ and $A = \mathbb{Z}$.

Example 0.6. : Let $A = \mathbb{Z}$ and $S = \{1, p, p^2, \ldots, p^n, \ldots\}$, then $S^{-1}A = \{\frac{m}{p^n} \mid m\mathbb{Z}, n \in \mathbb{N}\}$.

Example 0.7. : Let A be a ring and $S = \{0, 1\}$, then for $\forall \frac{a}{t} \in S^{-1}A$, we have $\frac{a}{t} = \frac{0}{1}$ (since $0 \cdot (a \cdot 1 - t \cdot 0) = 0$), the zero element. Thus $S^{-1}A = 0$; Conversely, if $S^{-1}A = 0$, then $\frac{1}{1} = \frac{0}{1}$ and therefore $\exists u \in S$ such that $u(1 \cdot 1 - 1 \cdot 0) = u = 0$, which means that $0 \in S$.

So we have proved that $S^{-1}A = 0 \Leftrightarrow 0 \in S$.

Example 0.8. : Let $A = \mathbb{Z}_6$ and $S = \{\overline{1}, \overline{3}\}$. Then for $\frac{\overline{2}}{\overline{1}} \in S^{-1}A$, we can know that $\frac{\overline{2}}{\overline{1}} = \frac{\overline{0}}{\overline{1}}$ since $\overline{3}(\overline{2} \cdot \overline{1} - \overline{1} \cdot \overline{0}) = \overline{0}$.

Now define a ring homomorphism

$$f : A \to S^{-1}A$$

$$a \mapsto \frac{a}{1}$$

note that f is in general not injective. (see example 0.8)

Proposition 0.2. : *Let $g : A \to B$ be a ring homomorphism, S is a multiplicative set of A, $g(s)$ is a unit of B for $\forall s \in S$. Then there exists a unique ring homomorphism $h : S^{-1}A \to B$ such that $h \circ f = g$, i.e., the diagram is commutative.*

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & B \\ S^{-1}A \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Proof: Define

$$h : S^{-1}A \to B$$

$$\frac{a}{s} \mapsto g(a)g(s)^{-1}$$

then h is a ring homomorphism and it's easy to check that h is well-defined. For $\forall a \in A$, $(h \circ f)(a) = h(f(a)) = h(\frac{a}{1}) = g(a)$, which means that $h \circ f = g$.

Let $h' : S^{-1}A \to B$ be any other homomorphism such that $h' \circ f = g$, then $h'(\frac{a}{s}) = h'(\frac{a}{1})h'(\frac{1}{s}) = h'(f(a))[h'(\frac{s}{1})]^{-1} = (h' \circ f)(a)[(h' \circ f)(s)]^{-1} = g(a)g(s)^{-1} = h(\frac{a}{s})$ $(\forall \frac{a}{s} \in S^{-1}A)$, hence $h' = h$. $\square$

For the ring homomorphism

$$f : A \to S^{-1}A$$

$$a \mapsto \frac{a}{1}$$

we can know that

(1) for $s \in S$, then $f(s)$ is a unit of $S^{-1}A$.

(2) if $f(a) = 0 \in S^{-1}A$, then $\exists u \in S$ such that $ua = 0$.

(3) every element of $S^{-1}A$ is of the form $f(a)f(s)^{-1}$.

In fact, we can characterize $S^{-1}A$ by these three conditions.

Proposition 0.3. *: Let $g : A \to B$ be a ring homomorphism such that*
(1) for $\forall s \in S$, $g(s)$ is unit in B;
(2) if $f(a) = 0$, then $\exists u \in S$ such that $ua = 0$;
(3) $\forall b \in B$, b is of the form $g(a)g(s)^{-1}$,
then there exists a unique isomorphism $h : A \to B$ such that $h \circ f = g$.

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & B \\ S^{-1}A \arrow[ur, "h"'] & \end{tikzcd}$$

Proof: Define

$$h : S^{-1}A \to B$$
$$\frac{a}{s} \mapsto g(a)g(s)^{-1},$$

then it's easy to check that h is well-defined and a ring homomorphism. By(3), h is clearly surjective. Suppose $\frac{a}{s} \in S^{-1}A$ such that $h(\frac{a}{s}) = g(a)g(s)^{-1} = 0$, hence $g(a) = 0$. By (2), $\exists u \in S$ such that $ua = 0$, i.e.,$u(a \cdot 1 - s \cdot 0) = 0$ and hence $\frac{a}{s} = 0$, which implies that $\ker h = 0$ and h is injective. Thus h is an isomorphism, $S^{-1}A \cong B$.

The uniqueness of h is similar to the proof of Proposition 0.2 $\square$

Let $\mathfrak{p}$ be a prime ideal of A and $S = A \setminus \mathfrak{p}$, then $1_A \in S$ and for $\forall a, b \in S$, we have $ab \in S$ (otherwise, if $ab \notin S = A \setminus \mathfrak{p}$, then $ab \in \mathfrak{p}$, which implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, a contradiction!), hence $S = A \setminus \mathfrak{p}$ is a multiplicative set of A.

Denote: $S^{-1}A = A_\mathfrak{p}$.

Example 0.9. : Let k be a field and $\mathfrak{p}$ is a prime ideal in $A = k[x_1, x_2, \ldots, x_n]$, let V be the variety of $\mathfrak{p}$ (i.e.,$V = \{(a_1, a_2, \ldots, a_n) \in k^n | \ f(a_1, a_2, \ldots, a_n) = 0 \text{ for } \forall f \in \mathfrak{p}\}$), then $A_\mathfrak{p} = \{\frac{f}{g} | \ g \notin \mathfrak{p}\}$. (note that $g \notin \mathfrak{p} \Leftrightarrow g$ does not vanish at any point of V)

Let $\mathfrak{p}$ be a prime ideal in a ring A, consider $\frac{x}{s} \in A_\mathfrak{p}$, if $x \in \mathfrak{p}$, then $\frac{x}{s}$ is not a unit. Indeed, if $\frac{x}{s}$ is a unit in $A_\mathfrak{p}$, then $\exists \frac{y}{t} \in A_\mathfrak{p}$ such that $\frac{x}{s} \cdot \frac{y}{t} = \frac{1}{1}$, i.e., $\exists u \in S = A \setminus \mathfrak{p}$ such that $u(xy - st) = 0$, hence $xyu = stu$, therefore $xyu = stu \notin \mathfrak{p}$, however, $x \in \mathfrak{p} \Rightarrow xyu \in \mathfrak{p}$, which is a contradiction;

Conversely, let $\frac{x}{s} \in A_\mathfrak{p}$ be a non-unit, if $x \notin \mathfrak{p}$, then $x \in S$ and $\frac{s}{x} \in A_\mathfrak{p}$, which implies $\frac{x}{s} \cdot \frac{s}{x} = \frac{1}{1}$ and hence $\frac{x}{s}$ is a unit (contradiction!).

Hence we have the following propostion

Proposition 0.4. *: Let A be a ring and $\mathfrak{p}$ is a prime ideal of A, then the set of non-units of $A_{\mathfrak{p}}$ is*

$$\mathfrak{p}A_{\mathfrak{p}} := \{\frac{x}{s} |\ x \in \mathfrak{p}, s \in S = A \setminus \mathfrak{p}\}.$$

Corollary 0.1. *: Let A be a ring and $\mathfrak{p}$ is a prime ideal of A, then $A_{\mathfrak{p}}$ is a local ring with the unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.*

Proof: Recall that if the set of all non-units forms an ideal of the ring A, then it must be its unique maximal ideal. □

Remark 0.1. *: In general, $S^{-1}A$ need not to be a local ring.*

Proposition 0.5. *: Let I be an ideal of A, S is a multiplicative set of A. Define $S^{-1}I := \{\frac{x}{s} |\ x \in I, s \in S\}$, then*

$$S^{-1}I = S^{-1}A \Leftrightarrow S \cap I \neq \phi$$

Proof: It's easy to see that $S^{-1}I$ is an ideal of $S^{-1}A$.

($\Leftarrow$): suppose $x \in I \cap S$, then $\frac{x}{x} = \frac{1}{1} \in S^{-1}I$, i.e., $S^{-1}I$ contains the identity of $S^{-1}A$, hence $S^{-1}I = S^{-1}A$;

($\Rightarrow$): since $S^{-1}I = S^{-1}A$, $\exists \frac{x}{s} \in S^{-1}I$ such that $\frac{x}{s} = \frac{1}{1}$, i.e., $\exists u \in S$ such that $u(x-s) = 0$, hence $xu = su \in S$. Thus $xu \in I \cap S$ and $S \cap I \neq \phi$. □

Now we talke about localization of modules:

Let S be a multiplicative set of A and M is an A-module. Define a relation $\sim$ on $M \times S$ by

$$(m_1, s_1) \sim (m_2, s_2) \Leftrightarrow \exists u \in S \text{ such that } u(m_1 s_2 - m_2 s_1) = 0$$

then $\sim$ is an equivalence relation on $M \times S$. The set of all equivalence classes is denoted by $S^{-1}M$ and $\frac{m}{s} \triangleq [(m, s)]$.

Define the addition on $S^{-1}M$ by

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} := \frac{s_2 m_1 + s_1 m_2}{s_1 s_2}$$

then $S^{-1}M$ is an abelian group with respect to this addition; Define the scalar multiplication

$$S^{-1}A \times S^{-1}M \to S^{-1}M$$

$$(\frac{a}{s}, \frac{b}{t}) \mapsto \frac{ab}{st}$$

then $S^{-1}M$ becomes an $S^{-1}A$-moudule w.r.t. this scalar multiplication.

Particularly, if $\mathfrak{p}$ is a prime ideal of A and $S = A \setminus \mathfrak{p}$, then $S^{-1}M$ is denoted by $M_{\mathfrak{p}}$.

Let $f: M \to N$ be an A-module homomorphism, define a map $S^{-1}f$ by

$$S^{-1}f : S^{-1}M \to S^{-1}N$$
$$\frac{m}{s} \mapsto \frac{f(m)}{s}$$

then it can be checked that $S^{-1}f$ is an $S^{-1}A$-module homomorphism.

Proposition 0.6. *: Let S be a multiplicative set of A, let*

$$M' \xrightarrow{f} M \xrightarrow{g} M''$$

be an exact sequence of A-modules, then

$$S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M''$$

is an exact sequence of $S^{-1}A$-modules.

Proof: Since $g \circ f : M' \to M''$, so we have $S^{-1}(g \circ f) : S^{-1}M' \to S^{-1}M''$. For $\forall \frac{m'}{s'} \in S^{-1}M'$, $(S^{-1}g \circ S^{-1}f)(\frac{m'}{s'}) = S^{-1}g(S^{-1}f(\frac{m'}{s'})) = S^{-1}g(\frac{f(m')}{s'}) = \frac{g(f(m'))}{s'} = S^{-1}(g \circ f)(\frac{m'}{s'})$, which means that $S^{-1}g \circ S^{-1}f = S^{-1}(g \circ f)$. Since $M' \xrightarrow{f} M \xrightarrow{g} M''$ is exact, so $\mathrm{Im} f = \ker g$ and $g \circ f = 0 : M' \to M''$. Thus by the definition we can know that $S^{-1}(g \circ f) = S^{-1}g \circ S^{-1}f = 0 : S^{-1}M' \to S^{-1}M''$, which implies that $\mathrm{Im}S^{-1}f \subseteq \ker S^{-1}g$; for $\forall \frac{m}{s} \in \ker S^{-1}g$, then $(S^{-1}g)(\frac{m}{s}) = \frac{g(m)}{s} = 0$, i.e., $\exists u \in S$ such that $u \cdot g(m) = 0$. Since $u \in S \subseteq A$ and g is an A-module homomorphism, therefore $ug(m) = g(um) = 0$, hence $um \in \ker g = \mathrm{Im} f$, thus $\exists m' \in M'$ such that $f(m') = um$, i.e., $\frac{m}{s} = \frac{f(m')}{us} = (S^{-1}f)(\frac{m'}{us}) \in \mathrm{Im}S^{-1}f$ and $\ker S^{-1}g \subseteq \mathrm{Im}S^{-1}f$. So we get $\ker S^{-1}g = \mathrm{Im}S^{-1}f$, the sequence is exact. □

Remark 0.2. *: By the proof above, we can know that S^{-1} is a covariant functor from the category of A-modules $\mathcal{M}_A$ to the category of $S^{-1}A$-modules $\mathcal{M}_{S^{-1}A}$.*

Proposition 0.7. *: Let S be a multiplicative set of the ring A, M is an A-module. Then we have*

$$S^{-1}A \otimes_A M \cong S^{-1}M.$$

Proof: Define

$$f' : S^{-1}A \times M \to S^{-1}M$$
$$(\frac{a}{s}, m) \mapsto \frac{am}{s}$$

then f' is an A-bilinear map. By the universal property of tensor product, $\exists$ a unique

$f \in Hom_A(S^{-1}A \otimes_A M, S^{-1}M)$ such that $f(\frac{a}{s} \otimes m) = f'(\frac{a}{s}, m) = \frac{as}{m}$.

$$\begin{tikzcd}
S^{-1}A \times M \arrow[r, "f'"] \arrow[d, "\otimes"'] & S^{-1}M \\
S^{-1}A \otimes_A M \arrow[ur, "f"'] &
\end{tikzcd}$$

it's clear that $f : S^{-1}A \otimes_A M \to S^{-1}M$ is surjective. For $\forall x \in S^{-1}A \otimes_A M$, then $x \triangleq \sum_{i=1}^{n} \frac{a_i}{s_i} \otimes m_i$, let $s = \prod_{i=1}^{n} s_i$ and let $t_i = \prod_{j \neq i} s_j$, then $x = \sum_{i=1}^{n} \frac{a_i t_i}{s} \otimes m_i = \sum_{i=1}^{n} \frac{1}{s} \otimes a_i t_i m_i = \frac{1}{s} \otimes (\sum_{i=1}^{n} a_i t_i m_i) \triangleq \frac{1}{s} \otimes m$, which means that any element of $S^{-1}A \otimes_A M$ is of the form $\frac{1}{s} \otimes m$, and the converse is true as well, i.e., $S^{-1}A \otimes_A M = \{\frac{1}{s} \otimes M |\ s \in S, m \in M\}$.

Suppose $\frac{1}{s} \otimes m \in \ker f$, hence $f(\frac{1}{s} \otimes m) = \frac{m}{s} = 0$, i.e., $\exists u \in S$ such that $u(m \cdot 1 - s \cdot 0) = um = 0$. Since $\frac{1}{s} = \frac{u}{us}$, we have $\frac{1}{s} \otimes m = \frac{u}{us} \otimes m = \frac{1}{us} \otimes (um) = \frac{1}{us} \otimes 0 = 0$, which implies $\ker f = \{0\}$ and f is injective.

Thus $f : S^{-1}A \otimes_A M \to S^{-1}M$ is an isomorphism and hence $S^{-1}A \otimes_A M \cong S^{-1}M$. $\square$